THU 12 NOV · 09:00–10:00 · PRIORY RD CMPLX F BLK 3F11 (SAME ROOM AS WEEK 7 — CHECK TIMETABLE)

Introduction to Pure Mathematics

MATH10027 · Group 07 · Permutations in cycle notation, calculating in dihedral groups, cancelling in a group

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THURSDAY 12 NOVEMBER 2026 · 09:00–10:00 · PRIORY RD CMPLX F BLK 3F11 (SAME ROOM AS WEEK 7 — CHECK TIMETABLE)

Introduction to Pure Mathematics — lesson at a glance

Permutations in cycle notation, calculating in dihedral groups, cancelling in a group.

Clock	What	Budget
09:00–09:05	Recall — 3 quick questions on last week, no notes	5 min
09:05–09:20	Round 1 — Permutations: from the array to cycles, and composing Watch: Pure PS7 A2 · Try: Pure PS7 A1	15 min
09:20–09:35	Round 2 — Dihedral groups: push every b to the right Watch: Example in D_{12} (hexagon), same method as PS7 A3 · Try: Pure PS7 A3	15 min
09:35–09:48	Round 3 — Reasoning in any group: cancel on the same side **(optional)** Watch: Pure PS7 A7(a) · Try: Pure PS7 A7(c)	13 min
09:48–09:55	Exit ticket — 3 questions alone, answers, “which round was hardest?”, reminders	7 min

Cut rule: Rounds 1 and 2 are the lesson. If Round 2 is not wrapped up by **09:36**, skip Round 3 completely, keep working on Round 2, and start the exit ticket at 09:48. Round 3 then becomes the first item of “At home”.

Opening words: “Two new families of groups this week: permutations and the symmetries of a polygon. Today is about doing, not watching: three short rounds. In each one I show one example, you do one like it with your neighbour, then we compare.”

> 开场 30 秒说清今天的玩法：三轮，每轮“我示范一道 → 你们两人做一道 → 一组说答案”。再用 30 秒说两个今天 13:00 的截止（PS6 指定题、Quiz 6 计分）。然后直接发 Recall。

BEFORE YOU START · tutor only

Deadlines and what has been lectured

- **Today 13:00:** Problem Sheet 6 Assigned Question closes (formative), and **Quiz 6** closes (**summative** — late = 0).
- **Problem Sheet 7 Assigned Question is ASSESSED**, due **13:00 Thursday 19 November**. In the room: method hints only (see the hints card near the bottom).
- **Quiz 7** (formative) opens today at 13:00. Part A solutions for Sheet 7 are already on Blackboard.

> **讲课进度（11 月 11 日查 Blackboard “Lecture summary”）：** 周一讲了 D_{2n} 的定义、$2n$ 个元素、标准形 a^ib^j、Cayley 表；周二讲了 S_n、两行记号、循环、不相交循环分解（7.21）。**“元素的阶”和“子群”（第 8 章）到周四早上还没讲**，所以今天主线不碰 A4–A6、A8、B1（11 月 8 日的旧草稿把它们排进了主线，这版已拿掉，放在页底答案库）。Tim 建议配这两讲的题正是 7.A1、A2、A3、A7——就是今天三轮的题。

TUTOR · HOW TO READ THE SYMBOLS ALOUD

Symbol reading table / 符号读法

Symbol	Read aloud	中文
S_n	“S n”, the symmetric group of degree n	n 次对称群
$(1,5,3,7)$	“the cycle one, five, three, seven”	循环：1→5→3→7→1
fg	“f g”: f after g (do g first)	先 g 后 f
$(fg)(k) = f(g(k))$	“f g of k equals f of g of k”	
D_{2n}	“D two n”, the dihedral group of order 2n	二面体群，2n 个元素
$a^i b^j$	“a to the i, b to the j”	标准形
a^{-1}	“a inverse”	逆元
$ba^i = a^{-i}b$	“b a to the i equals a to the minus i, b”	b 穿过 a 的幂，指数变号
$\Longleftrightarrow$	“if and only if”	当且仅当

PRINTED AT THE TOP OF THE HANDOUT

Key ideas (students have this box)

1. **Cycles.** $(1,5,3,7)$ means $1 \mapsto 5 \mapsto 3 \mapsto 7 \mapsto 1$; numbers not written stay fixed. *Recipe:* start at the smallest number, follow it until you return, close the bracket, restart at the smallest unused number.
2. **Compose right to left.** fg means “g first, then f”: $(fg)(k) = f(g(k))$. Usually $fg \neq gf$.
3. **Dihedral group.** $D_{2n} = \{e, a, \ldots, a^{n-1},\ b, ab, \ldots, a^{n-1}b\}$ with rules $a^n = e,\ \ b^2 = e,\ \ ba^i = a^{-i}b$.
4. **Cancel on the same side.** $xz = yz \Rightarrow x = y$ and $zx = zy \Rightarrow x = y$. Left undoes left, right undoes right.

RECALL · 09:00–09:05 · 5 min

Recall: three quick questions

No notes, 2 minutes on your own, then compare with your neighbour.

1. Which of these is **not** a group, and which axiom fails? $(\mathbb{Z}, +),\ (\mathbb{Z} \setminus \{0\}, \times),\ (\mathbb{R} \setminus \{0\}, \times)$.

   $(\mathbb{Z} \setminus \{0\}, \times)$: **inverses** fail — e.g. 2 has no inverse in $\mathbb{Z}$ (only ± 1 do). The other two are groups.

2. In a group G, solve $ax = b$ for x.

   $x = a^{-1}b$ — multiply by a^{-1} on the **left**.

3. True or false: in every group, $xy = yx$.

   False. Groups where it always holds are called abelian; permutations of $\{1,2,3\}$ do not all commute — we will see $fg \neq gf$ in Round 1.

> 2 分钟自己写、1 分钟和同桌对、2 分钟你报答案。不要展开讲——这只是把上周的“群公理、同侧消去”从记忆里取出来，第 3 轮要用。第 2 题有人写 ba^{-1} 就顺势问一句“左边还是右边？”，留个悬念到第 3 轮。

ROUND 1 · 09:05–09:20 · 15 min

Permutations: from the array to cycles, and composing

Skill: Write a permutation as disjoint cycles; compute fg right to left.

WATCH · 09:05–09:09 · you at the board

Pure PS7 A2

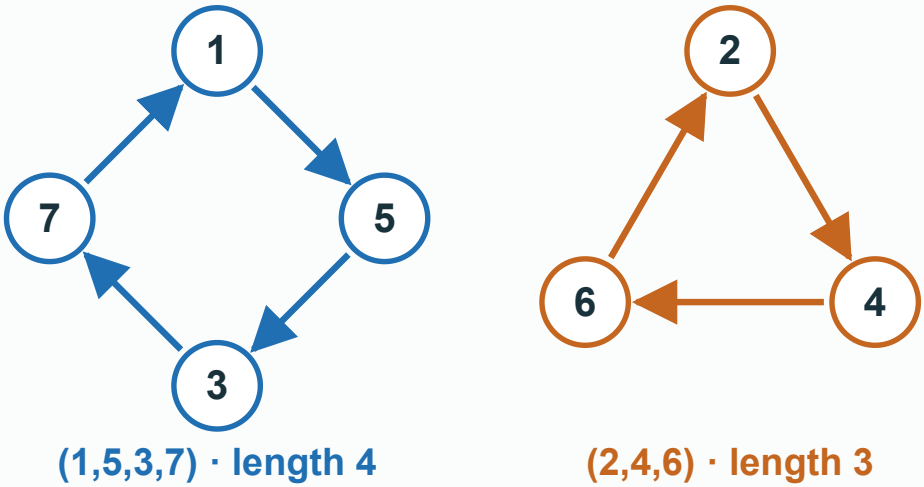
Express the following permutation as a product of disjoint cycles:

$$g = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 5 & 4 & 7 & 6 & 3 & 2 & 1 \end{pmatrix} \in S_7.$$

中文题意： 把 S_7 中用两行记号给出的置换 g 写成不相交循环的乘积。答案 $g = (1,5,3,7)(2,4,6)$。

Say: Follow one number at a time: where does 1 go, then where does that go, until you come back to 1. Close the bracket, then start again from the smallest number you have not used.

Board sketch / 板书图 · g = (1,5,3,7)(2,4,6): follow each number round its ring



Two disjoint cycles; nothing is fixed

Read column by column: $g(1) = 5,\ g(2) = 4,\ g(3) = 7,\ g(4) = 6,\ g(5) = 3,\ g(6) = 2,\ g(7) = 1.$

Start at 1: $1 \mapsto 5 \mapsto 3 \mapsto 7 \mapsto 1$, so the first cycle is $(1, 5, 3, 7)$.

Smallest unused number is 2: $2 \mapsto 4 \mapsto 6 \mapsto 2$, so the second cycle is $(2, 4, 6)$.

Every number $1, \ldots, 7$ has been used once, so

$$g = (1, 5, 3, 7)(2, 4, 6).$$

The cycles are disjoint, so the order does not matter: $g = (2, 4, 6)(1, 5, 3, 7)$ too.

4 分钟。先把两行记号抄上黑板，在旁边画箭头链 1→5→3→7→1（下面有现成的板书图）。只强调一个动作：“回到起点就关括号，再从最小的没用过的数开始”。不要在这里讲复合——留给学生在 Try 里自己撞上。

TRY · 09:09–09:15 · pairs, you walk round

Pure PS7 A1 · on the handout

Let

$$f = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 4 & 1 & 3 & 2 \end{pmatrix}, \qquad g = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 2 & 4 & 1 & 5 \end{pmatrix}.$$

Fill in the table (remember: g first, then f), then write fg as a product of disjoint cycles.

k	1	2	3	4	5
$g(k)$					
$f(g(k))$					

$fg =$

> **中文题意：**两个 S_5 中的置换，求 fg（先做 g 再做 f）的不相交循环分解。讲义上有一张三行的表让他们填：k、$g(k)$、$f(g(k))$。

While you walk round — say this, not the answer:

- Stuck at the start: “Where does g send 1? Now where does f send *that*?”
- Second row reads $5, 4, 1, 3, 2$: they applied f first. Ask “which one is next to the input k?”
- Wrote $(1)(3)(2, 4, 5)$: fine — say fixed points are normally left out.

Fast finishers (printed on the handout): Finished early? Compute gf as well. Is it the same as fg?

k	1	2	3	4	5
$g(k)$	3	2	4	1	5
$f(g(k))$	1	4	3	5	2

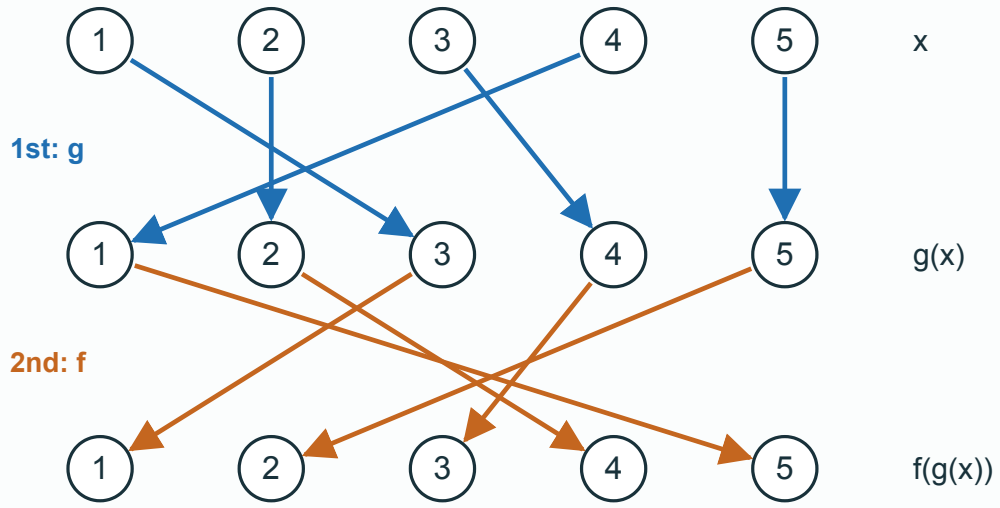
So 1 and 3 are fixed and $2 \mapsto 4 \mapsto 5 \mapsto 2$:

$$fg = (2, 4, 5).$$

Fast finishers: $(gf)(1) = g(5) = 5$, $(gf)(5) = g(2) = 2$, $(gf)(2) = g(4) = 1$, and $3, 4$ are fixed, so $gf = (1, 5, 2) \neq fg$.

In cycle notation $f = (1, 5, 2, 4, 3)$ and $g = (1, 3, 4)$, if anyone asks.

Board sketch / 板书图 · Three rows for fg: g first (blue), then f (orange)



fg = (2,4,5) (read top to bottom: g first, then f)

EXPLAIN · 09:15–09:20 · a pair reports

How: Ask one pair to read out their bottom row; write the table on the board; ask a second pair for the cycle form.

The trap: Going left to right gives $gf = (1, 5, 2)$ instead of $fg = (2, 4, 5)$. The function *next to the input* acts first.

Takeaway (students write it in their own words): fg = "g first, then f". Follow each number through; close the cycle when you get back to the start.

Quick check: Hands up: is $(5, 3, 7, 1)$ the same cycle as $(1, 5, 3, 7)$? Is $(1, 7, 3, 5)$?

$(5, 3, 7, 1)$: yes, same cycle, different starting point. $(1, 7, 3, 5)$: no — it runs backwards; it is the inverse.

5 分钟。请一组报表格最后一行，你只负责写。然后问全班"谁算出来的是 (1,5,2)？"——那是从左往右算的 gf，正好用来说"谁挨着输入谁先做"。最后让学生在讲义的 Takeaway 一栏自己写一句话。注意：PS7 计分指定题 (a) 也是求 fg，方法一样，但不要碰那道题的数字。

ROUND 2 · 09:20–09:35 · 15 min

Dihedral groups: push every b to the right

Skill: Reduce any product in D_{2n} to the standard form $a^i b^j$.

WATCH · 09:20–09:24 · you at the board

Example in D_{12} (hexagon), same method as PS7 A3

In $D_{12} = \{e, a, \ldots, a^5, b, ab, \ldots, a^5 b\}$ (the symmetries of a regular hexagon), write each of these in the form $a^i b^j$ with $0 \leqslant i \leqslant 5$, $j \in \{0, 1\}$:

(i) $ba^2\, ba^5$ (ii) $a^4\, ba^3$.

中文题意：在 D_{12}（正六边形的 12 个对称）里把两个乘积化成标准形 $a^i b^j$。答案：(i) a^3；(ii) ab。

Say: Three rules do everything: a to the six is e, b squared is e, and when b moves past a power of a, the power changes sign. So we push every b to the right and collect the a's.

Rules in D_{12} (box them at the top of the board):

$$a^6 = e, \qquad b^2 = e, \qquad ba^i = a^{-i} b.$$

$$\begin{aligned} \text{(i)} \quad ba^2\, ba^5 &= a^{-2} b\, ba^5 && (ba^2 = a^{-2} b) \\ &= a^{-2} a^5 && (b^2 = e) \\ &= a^3. \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad a^4\, ba^3 &= a^4\, a^{-3} b && (ba^3 = a^{-3} b) \\ &= ab. \end{aligned}$$

(i) has two b's and ends as a rotation a^3 ($j = 0$); (ii) has one b and ends as a reflection ab ($j = 1$).

4 分钟。先把三条规则框在黑板左上角并留着（Try 时学生要看）。每一行右边写用了哪条规则。用 D_{12} 而不是 D_{20} 示范，是为了把 A3 原题完整留给学生做。可以顺手画个六边形：a 转 60°，b 翻一下。

TRY · 09:24–09:30 · pairs, you walk round

Pure PS7 A3 · on the handout

Let $G = D_{20} = \{e, a, \ldots, a^9, b, ab, \ldots, a^9b\}$. Express the element

$$b^3a^5ba^{-7}$$

in the form a^ib^j with $i \in \{0, 1, \ldots, 9\}$ and $j \in \{0, 1\}$.

Rules in D_{20}: $a^{\square} = e$, $b^{\square} = e$, $ba^i = _____$

> **中文题意：**在 D_{20}（正十边形的 20 个对称）中，把 $b^3a^5ba^{-7}$ 化成 a^ib^j。讲义上先让他们自己填三条规则。答案 a^8。

While you walk round — say this, not the answer:

- Wrote $ba^5 = a^5b$: point at the boxed rule — “what happens to the exponent when b passes?”
- Stopped at a^{-12}: “the question wants i between 0 and 9 — which rule lets you add 10?”
- Treated b^3 as e: “what is b^2? so what is b^3?”

Fast finishers (printed on the handout): Finished early? Simplify $(a^3b)^2$ in D_{20}. What does the answer tell you about the symmetry a^3b?

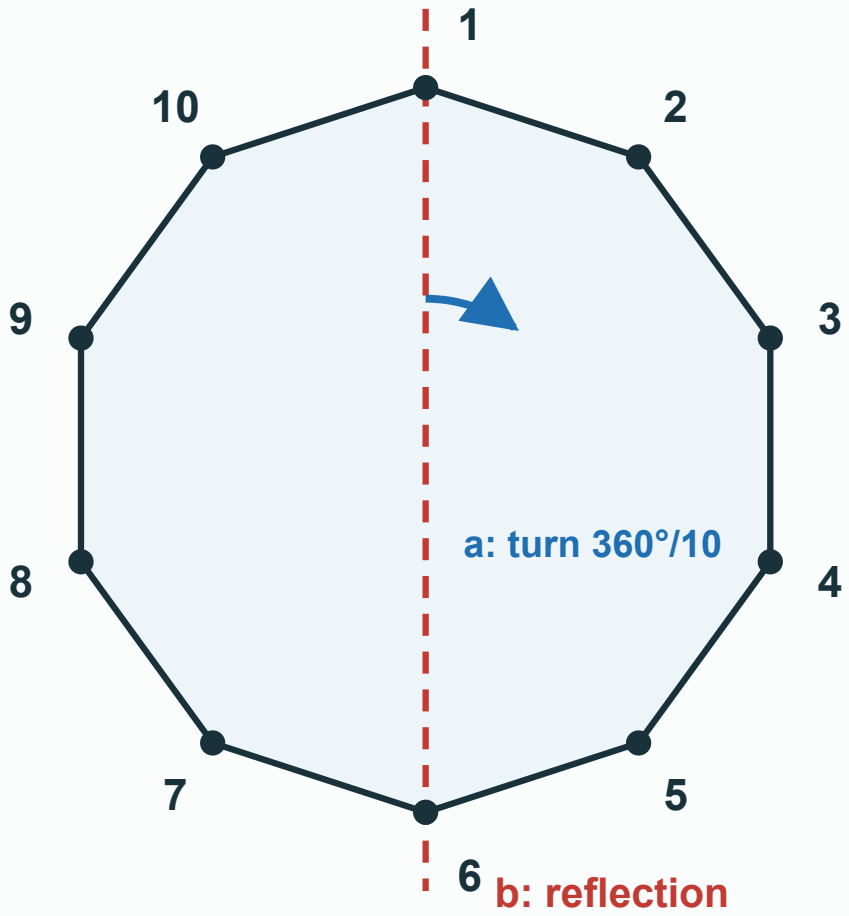
Rules: $a^{10} = e$, $b^2 = e$, $ba^i = a^{-i}b$.

$$\begin{aligned} b^3a^5ba^{-7} &= b\,a^5\,b\,a^{-7} && (b^3 = b^2b = b) \\ &= a^{-5}\,b\,b\,a^{-7} && (ba^5 = a^{-5}b) \\ &= a^{-5}a^{-7} = a^{-12} && (b^2 = e) \\ &= a^{-12}a^{20} = a^8 && (a^{10} = e). \end{aligned}$$

So $i = 8$, $j = 0$.

Fast finishers: $(a^3b)^2 = a^3\,ba^3\,b = a^3a^{-3}\,b\,b = e$. So a^3b undoes itself: it is a reflection. (Every a^ib is.)

Board sketch / 板书图 · D20: the regular 10-gon, rotation a and reflection b



Reflect, rotate, reflect = rotate the other way: ba = a^{-1}b

EXPLAIN · 09:30–09:35 · a pair reports

How: Ask a pair for their first line only, then a different pair for the next line, and so on — four pairs, four lines.

The trap: ba^5 is **not** a^5b: the exponent changes sign each time b passes. And do not stop at a^{-12}: reduce with $a^{10} = e$.

Takeaway (students write it in their own words): Push every b to the right; each time it passes a^i the exponent changes sign. Even number of b's → a rotation; odd → a reflection.

Quick check: Before anyone calculates: is $ab\,a^2\,b\,a\,b$ a rotation or a reflection?

Three b's (odd), so a reflection: the answer has $j = 1$. (In fact $ab\,a^2\,b\,ab = a\,a^{-2}\,b\,b\,ab = a^{-1}ab = b$.)

5 分钟。每组只报一行，四组接力，比你自己全写更能看出谁卡在哪。结尾用"数 b 的个数"这个检查：偶数个 b 是旋转，奇数个是反射——学生以后可以用它自查。

ROUND 3 · 09:35–09:48 · 13 min · OPTIONAL — skip if Round 2 runs past 09:36

Reasoning in any group: cancel on the same side

Skill: Solve an equation in a group; spot an illegal cancellation.

WATCH · 09:35–09:39 · you at the board

Pure PS7 A7(a)

Let G be a group. For fixed $a, b, c \in G$, prove that the equation $axb = c$ has a unique solution, and find it.

> **中文题意：**证明群里方程 $axb = c$ 有唯一解并求出来。答案 $x = a^{-1}cb^{-1}$。

> **Say:** In a group you can always undo a multiplication, but you must undo it on the same side. The a is on the left of x, so we remove it from the left; the b is on the right, so we remove it from the right.

$$\begin{aligned} axb = c &\iff a^{-1}(axb) = a^{-1}c && (\text{multiply on the left by } a^{-1}) \\ &\iff xb = a^{-1}c && (a^{-1}a = e) \\ &\iff (xb)b^{-1} = a^{-1}cb^{-1} && (\text{multiply on the right by } b^{-1}) \\ &\iff x = a^{-1}cb^{-1}. \end{aligned}$$

Each step can be undone, so $x = a^{-1}cb^{-1}$ is a solution and it is the only one. Check: $a(a^{-1}cb^{-1})b = c$.

> 4 分钟。板书时把“左乘”“右乘”两个词写出来。常见错误答案：$ca^{-1}b^{-1}$ 或 $a^{-1}b^{-1}c$——可以写在旁边问“为什么不行？”（因为群不一定交换）。这正好接上 Recall 第 2 题。

TRY · 09:39–09:45 · pairs, you walk round

Pure PS7 A7(c) · on the handout

A student argues: "If $xz = zy$, we can cancel z, so $x = y$."

(i) What is wrong with this argument?

(ii) Test it in S_3 with $x = (1,2)$, $y = (1,3)$, $z = (1,2,3)$: compute xz and zy.

> **中文题意：**有同学说"$xz = zy$ 可以消去 z 得 $x = y$"。(i) 错在哪？(ii) 在 S_3 里用给定的 x, y, z 验证：$xz = zy$ 但 $x \neq y$。原题要学生自己找反例；这里直接给出反例让他们验证，省时间，也顺便再练一次第 1 轮的复合。

While you walk round — say this, not the answer:

- For (i): "On which side of x is z? On which side of y?"
- For (ii): "Right to left again — in xz, which acts first?"

Fast finishers (printed on the handout): Finished early? Start A7(b): prove that $(xy)^2 = x^2y^2 \iff xy = yx$.

(i) We may only cancel z from the *same side*: $xz = yz \Rightarrow x = y$ and $zx = zy \Rightarrow x = y$. In $xz = zy$ the z is on the right of x but on the left of y. Multiplying on the right by z^{-1} only gives $x = zyz^{-1}$, which need not equal y because G need not be abelian.

(ii) Right to left:

$$xz = (1,2)(1,2,3): \quad 1 \mapsto 2 \mapsto 1, \;\; 2 \mapsto 3 \mapsto 3, \;\; 3 \mapsto 1 \mapsto 2, \quad \text{so } xz = (2,3).$$

$$zy = (1,2,3)(1,3): \quad 1 \mapsto 3 \mapsto 1, \;\; 2 \mapsto 2 \mapsto 3, \;\; 3 \mapsto 1 \mapsto 2, \quad \text{so } zy = (2,3).$$

So $xz = zy$ but $x = (1,2) \neq (1,3) = y$: the student's rule is false.

Fast finishers, A7(b): $(xy)^2 = x^2y^2 \iff xyxy = xxyy \iff yx = xy$, multiplying on the left by x^{-1} and on the right by y^{-1} (both steps reversible).

EXPLAIN · 09:45–09:48 · a pair reports

How: One sentence from a pair for (i); then both products from another pair.

> **The trap:** "Cancelling" is really multiplying by z^{-1} — and you have to say on which side.

> **Takeaway (students write it in their own words):** Undo on the same side: left with left, right with right. $xz = zy$ only gives $x = zyz^{-1}$.

> **Quick check:** In an abelian group, is the student right?
>
> Yes: then $zy = yz$, so $xz = yz$ and we can cancel on the right.

> 3 分钟。这一轮是可选的：第 2 轮 09:36 前没收完就整轮跳过，留作课后（A7 在 Blackboard 有公开答案）。

EXIT TICKET · 09:48–09:55 · 7 min

Exit ticket: three questions, on your own

1. Write $(1,2,3)(1,2)$ as a product of disjoint cycles.

   Right to left: $1 \mapsto 2 \mapsto 3$, $3 \mapsto 3 \mapsto 1$, $2 \mapsto 1 \mapsto 2$. Answer $(1,3)$.

2. In D_{2n}, simplify bab.

   $bab = a^{-1}b\,b = a^{-1}$ $(= a^{n-1})$.

3. In a group, solve $xa = b$ for x.

   $x = ba^{-1}$ — multiply by a^{-1} on the **right**.

Then: “Circle the round you are least sure about — hands up for Round 1? Round 2? Round 3?” Note the count; next week’s Recall comes from the round with most hands.

> 3 分钟各自写（不讨论），2 分钟你报答案、学生自己打勾。然后让他们在讲义上圈出“最没把握的一轮”，举手统计一下——哪一轮举手多，下周 Recall 就出那一轮的题。最后 1 分钟提醒截止：今天 13:00 PS6 指定题和 Quiz 6；PS7 指定题（计分）下周四 13:00。跳过了第 3 轮的话，第 3 题改成口头问。

After the tutorial (printed on page 1 of the handout)

- **From today:** finish A7(b); then B8 and C1 (they use only the group axioms).
- **After this week’s lectures on order and subgroups:** A4, A5, A6, A8, then B1.
- **Assigned Question (ASSESSED):** due 13:00 Thursday 19 November. Today’s Rounds 1 and 2 are the methods you need to start it.

Solutions to all Part A questions are on Blackboard now.

HAND-IN QUESTIONS · HINTS ONLY

Assigned questions: what you may and may not say

Assigned Question — this is an ASSESSED homework. Submit your solutions by **13:00 on Thursday November 19th**.

(a) Let $G = S_8$ be the symmetric group of degree 8 and consider the elements

$$f = (1,3,5)(2,6,7,8), \qquad g = (6,5,4,3,2).$$

Write fg as a product of disjoint cycles and find its order.

(b) Let $G = D_6$ be the dihedral group of order 6. Find two different subgroups of G with order 2.

(c) Let $G = D_{12}$ be the dihedral group of order 12. Find two different subgroups of G with order 6. Please explain your answer.

(d) Let G be a group. Prove that $|x| = |gxg^{-1}|$ for all $g, x \in G$.
[Hint. Show that if $n \in \mathbb{Z}^+$, then $x^n = e$ if and only if $(gxg^{-1})^n = e$.]

Hints only — do not give answers, cycles or subgroups.

- **(a)** Compose right to left: $(fg)(k) = f(g(k))$, so apply g first. Track each of $1, \ldots, 8$ (including points that g fixes) and close up each cycle. Only once fg is written as a product of *disjoint* cycles may you read off its order from the cycle lengths. A complete answer shows the tracking (or a table of values), the disjoint-cycle form and a one-line justification of the order.
- **(b)** A subgroup of order 2 must be $\{e, x\}$ for some x. Which x make such a set closed under products and inverses? List the six elements of D_6 with the lecture notation and decide which ones qualify. A complete answer names two different subgroups and says why each is a subgroup.
- **(c)** Look for subsets of size 6 that are described by a property — a type of symmetry, or the symmetries that preserve some extra structure you can draw on the hexagon — and then check the subgroup conditions. “Please explain” means: list the elements, justify closure, identity and inverses (or use a criterion from lectures), and explain why the two subgroups differ.
- **(d)** Follow the hint: expand $(gxg^{-1})^n$ for $n = 2, 3$ and look for cancellation; then prove the general statement for all $n \in \mathbb{Z}^+$ (induction is natural). Prove *both* directions of the “if and only if”, then explain carefully why this gives equal orders, including the case where x has infinite order.

Remind students: complete sentences, define notation, and show working.

> **计分作业（PS7 Assigned Question，11 月 19 日周四 13:00 截止）：课上绝对不给结论、不写出 (a) 的循环分解、不点名 (b)(c) 的具体子群、不写 (d) 的完整证明。**学生问“对不对”时一律反问：“你从右往左算了吗？”“这个集合对乘法封闭吗？”可以讲的通用方法：A1 的复合方法（Part A 有公开答案）、子群三条件、A8/B5 的判别法。常见失分点（只作提醒）：(a) 从左往右复合，或在非不交循环上直接取 lcm；(c) 没有解释；(d) 只证一个方向或漏掉无限阶的情形。B6、C2 课上讲时注意别顺带讲出本题答案。

Assigned Question (submit by 13:00 today, Thursday 12 November).

Which of the following are groups, and why? For those that are groups, find the identity element, find the inverse of every element, and decide whether or not the group is abelian. Please explain your answers.

(a) The set of even integers under addition.

(b) The set of odd integers under addition.

(c) The set $\mathbb{R}_+$ of positive real numbers under multiplication.

(d) $(\mathbb{Z}, \star)$, where $x \star y = x + y + 1$ for all $x, y \in \mathbb{Z}$.

(e) $(\mathbb{Z}, \bullet)$, where $x \bullet y = y$ for all $x, y \in \mathbb{Z}$.

Hand-in question: hints only, no verdicts. Use the definition of a group from the lecture notes once it has been covered. For each part, test in this order (write $*$ for the operation):

1. **Closure** (is $*$ a binary operation?): for all x, y in the set, is $x * y$ again in the set?
2. **Associativity:** is $(x * y) * z = x * (y * z)$ for all x, y, z? Expand both sides.
3. **Identity:** is there one element e in the set with $e * x = x$ *and* $x * e = x$ for every x?
4. **Inverses:** for each x, is there y *in the set* with $x * y = y * x = e$?
5. If it is a group: **abelian** means $x * y = y * x$ for all x, y.

Where to look in each part

- (a) Write even integers as $2a$ with $a \in \mathbb{Z}$ and check closure; then ask whether the natural identity candidate and the natural inverse candidate are even.
- (b) Start with closure: write odd integers as $2a + 1$ and compute the sum of two of them. Also ask whether the identity candidate is in the set.
- (c) Closure under multiplication; identity candidate 1; inverse candidate $1/x$. Check that each candidate lies in $\mathbb{R}_+$.
- (d) Expand $(x \star y) \star z$ and $x \star (y \star z)$ and compare. Find an identity candidate by solving $x \star e = x$ for e, then check $e \star x = x$ too. Find an inverse candidate by solving $x \star y = e$ for y and check that $y \in \mathbb{Z}$. Finally compare $x \star y$ with $y \star x$.
- (e) Check closure and associativity by expanding both sides. Then concentrate on the identity axiom: an identity must work on *both* sides, so test $e \bullet x = x$ and $x \bullet e = x$ separately, for all x.

A complete answer contains: a clear verdict for every part; if it is a group, the identity element, the inverse of every element (a formula), and abelian or not with a reason; if it is not a group, the axiom that fails together with an explicit counterexample using actual numbers.

> 交作业题，今天 13:00 截止：只给检查清单，不给结论，不当场判对错。非计分，只给反馈。群的定义第 7 章已讲，以讲义版本为准。学生追问某小题是不是群时，只反问：单位元要两边都成立吗？逆元在不在这个集合里？

LOOKUP · not part of today's route

Not yet lectured on Thursday morning: order and subgroups (PS7 A4–A6, A8, B1)

Use only if a student asks. Order of an element and subgroups are Chapter 8, which starts in Thursday's and Friday's lectures.

A4. What is the order of the element $g \in S_7$ in Question A2?

A6. Find an element $x \in S_{10}$ with $|x| = 21$.

Definition. The order $|x|$ of an element x is the smallest positive integer n with $x^n = e$ (and $|x| = \infty$ if there is none).

A4. From P1, $g = (1, 5, 3, 7)(2, 4, 6)$, a product of disjoint cycles $c_1 = (1, 5, 3, 7)$ of length 4 and $c_2 = (2, 4, 6)$ of length 3. Disjoint cycles commute, so $g^n = c_1^n c_2^n$. A k-cycle returns everything to its start exactly when n is a multiple of k. Since c_1 and c_2 move different numbers, $g^n = e$ if and only if $c_1^n = e$ and $c_2^n = e$, i.e. if and only if $4 \mid n$ and $3 \mid n$. The smallest such positive n is

$$|g| = \operatorname{lcm}(4, 3) = 12.$$

A6. We want cycle lengths with lowest common multiple $21 = 3 \cdot 7$ that fit into $\{1, \ldots, 10\}$. A single 21-cycle is impossible (only 10 numbers), but a 7-cycle and a disjoint 3-cycle use $7 + 3 = 10$ numbers:

$$x = (1, 2, 3, 4, 5, 6, 7)(8, 9, 10) \in S_{10}, \qquad |x| = \operatorname{lcm}(7, 3) = 21.$$

若第 8 章“元素的阶”还没讲，先写一行定义：|x| = 使 $x^n = e$ 的最小正整数 n。板书：P1 的两个循环各画一圈，问“转几次回原位”——4 和 3，一起回原位要 lcm(4,3) = 12。常见错误：把阶当成循环长度之和（7）或乘积（碰巧 12，换 (1,2)(3,4) 就错：lcm = 2，不是 4）；A6 写一个 21-循环（S_{10} 里不可能）。追问：S_{10} 里有阶 30 的元素吗？（需要 2+3+5 = 10：(1,2)(3,4,5)(6,7,8,9,10)，有。）

A5. How many elements in $G = D_{12}$ have order 4?

D_{12} is the group of symmetries of a regular hexagon: $G = \{e, a, \dots, a^5, b, ab, \dots, a^5b\}$ with $a^6 = b^2 = e$ and $ba^i = a^{-i}b$. Here a is rotation by $2\pi/6$.

Reflections. For every i, $(a^ib)^2 = a^i\,(ba^i)\,b = a^ia^{-i}b\,b = b^2 = e$, and $a^ib \neq e$, so each of the six elements a^ib has order 2.

Rotations. a^i is rotation by $i \cdot 60^\circ$. Count how many times we must apply it to get back to the start:

element	e	a	a^2	a^3	a^4	a^5
angle	0°	60°	120°	180°	240°	300°
order	1	6	3	2	3	6

For instance $(a^2)^3 = a^6 = e$ while $(a^2)^1, (a^2)^2 = a^4 \neq e$, so $|a^2| = 3$; and $(a^4)^3 = a^{12} = e$ while $a^4, a^8 = a^2 \neq e$, so $|a^4| = 3$.

The orders occurring are $1, 2, 3, 6$. **No element of D_{12} has order 4**, so the answer is 0.

板书画正六边形（见板书图），标 a 为逆时针转 60°、b 为一条对称轴。左栏 6 个转动配角度和阶，右栏 6 个反射都是阶 2。常见错误：以为 12 个元素里总有阶 4 的（4 整除 12 但不保证存在）；把 a^4 的阶说成 4。追问：D_{12} 里阶为 2 的元素有几个？（6 个反射加 a^3，共 7 个。）注意：PS7 指定题 (b)(c) 问 D_6、D_{12} 的子群，这里只讲元素的阶，不要顺势列子群。

A8. Let $G = (\mathbb{Z}, +)$ be the group of integers under addition. Let $H = 2\mathbb{Z}$ be the subgroup consisting of all the even integers, and let $K = 3\mathbb{Z}$ be the subgroup consisting of all the integers divisible by 3.

Prove that $H \cup K$ is not a subgroup of G.

A subset of G is a subgroup when it contains the identity, is closed under the operation and is closed under inverses. To show something is *not* a subgroup, it is enough to show that one of these fails, with explicit elements.

We have $2 \in H$ and $3 \in K$, so $2, 3 \in H \cup K$. But

$$2 + 3 = 5,$$

and 5 is neither even nor divisible by 3, so $5 \notin H$ and $5 \notin K$, i.e. $5 \notin H \cup K$. Hence $H \cup K$ is not closed under the group operation $+$, and therefore $H \cup K$ is not a subgroup of G.

Remark. The other two conditions do hold: $0 \in H \cup K$, and if $n \in H \cup K$ then $-n \in H \cup K$. Closure is the only thing that breaks. (Problem B4 shows this is typical: $H \cup K$ is a subgroup only when one of H, K contains the other.)

若第 8 章“子群”还没讲，先写一行：子群 = 含单位元、对运算封闭、对逆封闭的子集。提醒这里运算是加法：“乘积”就是和，单位元是 0，逆是 −n。板书数轴上标出 H、K 的点（偶数、3 的倍数），2 和 3 都在并集里，5 不在。常见错误：只说“看起来不封闭”而不给具体数字；去检查单位元和逆（它们其实成立）。追问：H ∩ K 是子群吗？（是，$6\mathbb{Z}$。）

B1. Determine the set $\{|x| : x \in S_8\}$ of orders of elements in S_8. Justify your answer.

Key fact (P4). Write $x \in S_8$ as a product of disjoint cycles of lengths $\ell_1, \dots, \ell_k$, including fixed points as cycles of length 1, so $\ell_1 + \cdots + \ell_k = 8$. Then $|x| = \operatorname{lcm}(\ell_1, \dots, \ell_k)$. Conversely, every list of positive integers with sum 8 is the list of cycle lengths of some $x \in S_8$ (fill in $1, 2, \dots, 8$ in order).

So the possible orders are exactly the numbers $\operatorname{lcm}(\ell_1, \ldots, \ell_k)$ for lists with $\ell_1 \geqslant \cdots \geqslant \ell_k \geqslant 1$ and $\ell_1 + \cdots + \ell_k = 8$. There are 22 such lists; organise them by the longest cycle:

cycle lengths	example x	$\lvert x\rvert$
[8]	$(1,2,3,4,5,6,7,8)$	8
[7,1]	$(1,2,3,4,5,6,7)$	7
[6,2]	$(1,2,3,4,5,6)(7,8)$	6
[6,1,1]	$(1,2,3,4,5,6)$	6
[5,3]	$(1,2,3,4,5)(6,7,8)$	15
[5,2,1]	$(1,2,3,4,5)(6,7)$	10
[5,1,1,1]	$(1,2,3,4,5)$	5
[4,4]	$(1,2,3,4)(5,6,7,8)$	4
[4,3,1]	$(1,2,3,4)(5,6,7)$	12
[4,2,2]	$(1,2,3,4)(5,6)(7,8)$	4
[4,2,1,1]	$(1,2,3,4)(5,6)$	4
[4,1^4]	$(1,2,3,4)$	4
[3,3,2]	$(1,2,3)(4,5,6)(7,8)$	6
[3,3,1,1]	$(1,2,3)(4,5,6)$	3
[3,2,2,1]	$(1,2,3)(4,5)(6,7)$	6
[3,2,1^3]	$(1,2,3)(4,5)$	6
[3,1^5]	$(1,2,3)$	3
[2,2,2,2]	$(1,2)(3,4)(5,6)(7,8)$	2
[2,2,2,1,1]	$(1,2)(3,4)(5,6)$	2
[2,2,1^4]	$(1,2)(3,4)$	2
[2,1^6]	$(1,2)$	2
[1^8]	e	1

Reading off the last column,

$$\{|x| : x \in S_8\} = \{1,2,3,4,5,6,7,8,10,12,15\}.$$

Why nothing else: every element has one of the 22 cycle types above, so its order is in this set. For instance 9 would need a 9-cycle (too long); $14 = \operatorname{lcm}(7,2)$ needs $7+2=9>8$ points; 20 needs $5+4=9$; 24 needs $8+3$; 30 needs $5+3+2=10$.

> 板书按最长循环分组列 22 种长度组合（可先让学生分组：一组做最长 8–5，一组做最长 4–1），每行只写长度和 lcm，例子口头带过。结论 {1,2,3,4,5,6,7,8,10,12,15}。常见错误：漏掉 15（5+3）和 12（4+3+1）；把 9、14 当成

> 可能。追问：S_8 中最大的阶是多少？（15。）官方解答里这张表印在了 B2 下面，应属于 B1。时间紧：只列能产生“新”阶的 6 行（8, 7, 6, 5+3, 5+2, 4+3），其余口头说只给出已有的阶。

LOOKUP · not part of today's route

Rest of Sheet 7, Parts B and C

Lookup only.

(a) Determine $\{|x| : x \in D_{20}\}$.

(b) Let G be the symmetry group of a cube. Describe (geometrically) an element in G of order 2, an element of order 3, and an element of order 4.

(a) Write $D_{20} = \{e, a, \ldots, a^9, b, ab, \ldots, a^9b\}$ with $a^{10} = b^2 = e$ and $ba = a^{-1}b$, so $|a| = 10$.

Rotations. For $0 \leqslant i \leqslant 9$ and $k \in \mathbb{Z}^+$, $(a^i)^k = a^{ik} = e$ if and only if $10 \mid ik$, which happens if and only if $\frac{10}{\gcd(10,i)} \mid k$. Hence $|a^i| = \frac{10}{\gcd(10,i)}$:

$$|e| = 1, \quad |a| = |a^3| = |a^7| = |a^9| = 10, \quad |a^2| = |a^4| = |a^6| = |a^8| = 5, \quad |a^5| = 2.$$

Reflections. Since $ba^i = a^{-i}b$, we get $(a^ib)^2 = a^i(ba^i)b = a^ia^{-i}b^2 = e$, and $a^ib \neq e$. So each of the ten elements a^ib has order 2.

Therefore $\{|x| : x \in D_{20}\} = \{1, 2, 5, 10\}$.

(b)

- **Order 2:** the reflection in the horizontal plane through the middle of the cube (or the rotation through π about the axis through the centres of two opposite faces). Doing it twice is the identity.
- **Order 3:** the rotation through $2\pi/3$ about the axis through two diagonally opposite corners. It permutes the three edges at the top corner cyclically, so the first time it is the identity is after three turns.
- **Order 4:** the rotation through $\pi/2$ about the axis through the centres of two opposite faces.

> 板书先写 D_{20} 的元素表和关系 $a^{10} = b^2 = e$, $ba = a^{-1}b$，再列一行 $i = 0, \ldots, 9$ 对应 $|a^i|$。常见错误：把 a^ib 的阶也按旋转算；或者只给答案集合不说明理由。追问：D_{12} 里为什么没有 4 阶元？（接 A5）。注意：sol7 把 B1 的 S_8 阶表（Table 1）印在了 B2(a) 之后，那张表属于 B1，不属于本题。立方体部分口头描述即可，时间紧只讲 (a)。

(a) Let G be an abelian group and let $x, y \in G$ be elements of finite order, where $|x| = m$ and $|y| = n$.

Show that $(xy)^k = e$ for every positive integer k that is divisible by both m and n. Deduce that $|xy|$ divides $\mathrm{lcm}(m, n)$, which is the lowest common multiple of m and n.

(b) Give an example of a non-abelian group G together with a pair of finite order elements $x, y \in G$ such that $|xy|$ does not divide $\mathrm{lcm}(|x|, |y|)$.

(a) Since G is abelian, $(xy)^k = x^ky^k$ for every $k \in \mathbb{Z}^+$ (rearrange the k copies of x to the front; formally by induction: $(xy)^{k+1} = (xy)^kxy = x^ky^kxy = x^{k+1}y^{k+1}$).

Let k be a positive integer divisible by m and by n, say $k = sm = tn$ with $s, t \in \mathbb{Z}^+$. Then

$$(xy)^k = x^ky^k = (x^m)^s(y^n)^t = e^se^t = e.$$

In particular $k = \mathrm{lcm}(m, n)$ gives $(xy)^{\mathrm{lcm}(m,n)} = e$. So xy has finite order, and by Proposition 8.7(a) (if $g^k = e$ then $|g|$ divides k) we conclude $|xy| \mid \mathrm{lcm}(m, n)$.

(b) Take $G = D_6 = \{e, a, a^2, b, ab, a^2b\}$, the symmetries of an equilateral triangle, and $x = ab$, $y = b$. Both are reflections, so $|x| = |y| = 2$ and $\mathrm{lcm}(2, 2) = 2$. But

$$xy = ab\,b = ab^2 = a,$$

which has order 3, and $3 \nmid 2$. (Equivalently in S_3: $x = (1,2)$, $y = (1,3)$ give $xy = (1,3,2)$ of order 3.)

> 板书 (a) 第一行就写 $(xy)^k = x^k y^k$ 并标注“需要 abelian”，这是全题唯一用到交换律的地方。常见错误：以为 $|xy| = \mathrm{lcm}(m,n)$（只能说整除，反例 $y = x^{-1}$ 时 $xy = e$）；(b) 举的例子没有算出 xy 的阶。追问：在 S_3 里找同样的例子（两个对换乘起来是 3-循环）。时间紧只讲 (b)，两行就够。

Let H and K be subgroups of a group G. Prove that $H \cup K$ is a subgroup of G if and only if either $H \leqslant K$ or $K \leqslant H$.

($\Leftarrow$) If $H \leqslant K$ then $H \cup K = K$, which is a subgroup. Similarly, if $K \leqslant H$ then $H \cup K = H$ is a subgroup.

($\Rightarrow$) We prove the contrapositive: if $H \not\subseteq K$ and $K \not\subseteq H$, then $H \cup K$ is not a subgroup.

So assume there are elements $h \in H \setminus K$ and $k \in K \setminus H$. Consider $hk \in G$.

- If $hk \in H$, then $k = h^{-1}(hk) \in H$, because H is closed under inverses and products. This contradicts $k \notin H$.
- If $hk \in K$, then $h = (hk)k^{-1} \in K$, because K is closed under inverses and products. This contradicts $h \notin K$.

Hence $hk \notin H \cup K$, although $h, k \in H \cup K$. So $H \cup K$ is not closed under products and is not a subgroup of G.

> 接在 A8（$2\mathbb{Z} \cup 3\mathbb{Z}$）之后讲最自然：A8 的 $2 + 3 = 5$ 正是这里 hk 的具体例子。板书先写“逆否命题”一行，再分两种情况。常见错误：只证一个方向；或在情况 1 里写成 $k = (hk)h^{-1}$（顺序错，非交换群里不成立）。追问：三个子群的并呢？（可以是子群，例如 Klein 四元群是三个 2 阶子群的并——只作拓展，不展开）。时间紧：只讲 ($\Rightarrow$) 的两种情况。

Let H be a non-empty subset of a group G. Prove that H is a subgroup of G if $x^{-1}y \in H$ for all $x, y \in H$.

We must show that H contains e, is closed under inverses and is closed under products. The order of the three steps matters.

1. **Identity.** H is non-empty, so pick $h \in H$. Taking $x = y = h$ gives $e = h^{-1}h = x^{-1}y \in H$.
2. **Inverses.** Let $x \in H$. Since $e \in H$ by step 1, taking $y = e$ gives $x^{-1} = x^{-1}e \in H$.
3. **Products.** Let $g, h \in H$. By step 2, $g^{-1} \in H$. Taking $x = g^{-1}$ and $y = h$ gives $x^{-1}y = (g^{-1})^{-1}h = gh \in H$.

Associativity is inherited from G. Hence $H \leqslant G$.

> 板书三步编号写清楚，强调顺序：没有第 1 步就不能在第 2 步取 $y = e$，没有第 2 步就不能在第 3 步取 $x = g^{-1}$。常见错误：忘了用“非空”；第 3 步直接取 $x = g$（得到的是 $g^{-1}h$，不是 gh）。追问：为什么必须要求非空？（空集满足条件但不是子群）。这一判别法 C4(e) 和 B6 的另解都会用到。

Let G be a group, let H be a subgroup and fix an element $x \in G$. Define

$$xHx^{-1} = \{xhx^{-1} : h \in H\}.$$

Prove that xHx^{-1} is a subgroup of G and $|H| = |xHx^{-1}|$.

Write $K = xHx^{-1}$. Since $e \in H$, K is non-empty.

Subgroup. We use B5. Let xax^{-1}, $xbx^{-1} \in K$ with $a, b \in H$. Then

$$(xax^{-1})^{-1}(xbx^{-1}) = (xa^{-1}x^{-1})(xbx^{-1}) = xa^{-1}(x^{-1}x)bx^{-1} = x(a^{-1}b)x^{-1}.$$

Since H is a subgroup, $a^{-1}b \in H$, so this element lies in K. By B5, $K \leqslant G$. (Checking identity $xex^{-1} = e$, closure $(xax^{-1})(xbx^{-1}) = x(ab)x^{-1}$ and inverses $(xhx^{-1})^{-1} = xh^{-1}x^{-1}$ separately also works.)

Same size. Define $\varphi : H \to K$ by $\varphi(h) = xhx^{-1}$. By the definition of K, φ is surjective. It is injective: if $\varphi(a) = \varphi(b)$, then $xax^{-1} = xbx^{-1}$; multiplying on the left by x^{-1} and on the right by x gives $a = b$. So φ is a bijection and $|H| = |K|$.

> 板书先写 K 的元素"长相" $x \square x^{-1}$，再做一次 B5 检验，让学生看到中间 $x^{-1}x$ 抵消。$|H| = |K|$ 用双射 $h \mapsto xhx^{-1}$ 和左右消去律。常见错误：写 $(xhx^{-1})^{-1} = x^{-1}h^{-1}x$（逆的顺序错）。**注意：本题手法与 PS7 计分作业 Assigned Question (d)（$|x| = |gxg^{-1}|$）很接近。课上可以放开讲"子群"和"双射"两部分，但不要讲 $(gxg^{-1})^n = gx^ng^{-1}$ 以及由此推出元素的阶不变的论证。**

Find a subgroup of $G = S_6$ of order 48.

[Hint. Consider the stabiliser

$$H = \{g \in G : g(a) \in A \text{ for all } a \in A\}$$

of a suitably defined subset A of $\{1, 2, 3, 4, 5, 6\}$.]

Let $X = \{1, \ldots, 6\}$, let $A \subseteq X$ and let H be as in the hint.

H is a subgroup.

- $e(a) = a \in A$ for all $a \in A$, so $e \in H$.
- If $f, g \in H$ and $a \in A$, then $g(a) \in A$, so $(fg)(a) = f(g(a)) \in A$. Thus $fg \in H$.
- Let $g \in H$. Then g maps A into A and is injective, so (as A is finite) its restriction is a bijection $A \to A$. Hence for $a \in A$ there is $a' \in A$ with $g(a') = a$, i.e. $g^{-1}(a) = a' \in A$. So $g^{-1} \in H$.

Counting. If $|A| = m$, every $g \in H$ permutes A and therefore also permutes $X \setminus A$ (it is a bijection of X mapping A onto A). Conversely, any permutation f_1 of A together with any permutation f_2 of $X \setminus A$ defines an element of H. So $|H| = m!\,(6 - m)!$.

Choice. We need $m!(6 - m)! = 48 = 2! \cdot 4!$. Take $A = \{1, 2\}$: then $H = \{g \in S_6 : g(\{1, 2\}) = \{1, 2\}\}$ is a subgroup of S_6 of order 48.

> 板书先画 $\{1, \ldots, 6\}$ 分成 A 和 $X \setminus A$ 两块，再写计数 $m!(6 - m)!$，最后解 $m!(6 - m)! = 48$（$m = 2$ 或 4）。常见错误：只验证 $|H| = 48$ 而不证 H 是子群；逆元那一步没用到"A 有限、单射即双射"。追问：$m = 3$ 时阶是多少？（36）。时间紧：子群部分只讲逆元那一条，其余口头带过。

Let S be a set with a binary operation $\star$ satisfying all the group axioms, except for the existence of inverses. In particular, $\star$ is associative, and S has an identity element e such that $e \star a = a = a \star e$ for all $a \in S$.

(a) Let

$$G = \{a \in S : a \star x = x \star a = e \text{ for some } x \in S\}$$

be the set of elements in S with an inverse in S. Prove that $(G, \star)$ is a group.

[Hint. First check that the restriction of $\star$ to $G \times G$ defines a binary operation on G.]

(b) Give some examples of groups you have seen that arise naturally in this way.

Write $\star$ multiplicatively.

(a) *Binary operation on G.* Let $a, b \in G$, with $ax = xa = e$ and $by = yb = e$ for some $x, y \in S$. Then

$$(ab)(yx) = a(by)x = aex = ax = e, \qquad (yx)(ab) = y(xa)b = yeb = yb = e,$$

so $ab \in G$. Hence $\star$ restricts to a binary operation $G \times G \to G$.

Associativity holds in G because it holds in S.

Identity. $ee = e$, so $e \in G$, and e is an identity for G.

Inverses. If $a \in G$ with $ax = xa = e$, then the same equations show $x \in G$ (its inverse in S is a), and x is an inverse of a in G.

So $(G, \star)$ is a group.

(b) Examples:

- $(\mathbb{R} \setminus \{0\}, \times)$ inside $(\mathbb{R}, \times)$;
- $(\{1, -1\}, \times)$ inside $(\mathbb{Z}, \times)$;
- $GL_n(\mathbb{R})$, the invertible $n \times n$ real matrices, inside $(M_n(\mathbb{R}), \times)$;
- the permutations of a set X (bijections $X \to X$) inside all functions $X \to X$ under composition.

> 给学得快的同学。板书核心一行：$(ab)(yx) = a(by)x = e$，强调逆的顺序要反过来（和"先穿袜子后穿鞋"同理）。常见错误：只验证 $(ab)(yx) = e$，忘了另一边 $(yx)(ab) = e$；忘了说明 $x \in G$。(b) 让学生自己举例，$GL_n(\mathbb{R})$ 和置换群最有代表性。

Let G be a group such that $x^2 = e$ for all $x \in G$. Prove that G is abelian.

Let $x, y \in G$. Applying the hypothesis to xy, x and y:

$$(xy)^2 = e = ee = x^2y^2.$$

By A7(b), $(xy)^2 = x^2y^2$ if and only if $xy = yx$. Hence $xy = yx$ for all $x, y \in G$, so G is abelian.

Alternative. $x^2 = e$ means $x = x^{-1}$ for every x. So $xy = (xy)^{-1} = y^{-1}x^{-1} = yx$.

> 两行证明，适合提前做完的学生。板书写第二种证法更直观：$xy = (xy)^{-1} = y^{-1}x^{-1} = yx$。常见错误：写 $(xy)^{-1} = x^{-1}y^{-1}$（顺序错）。追问：举一个满足条件的非平凡群（如 $\mathbb{Z}_2 \times \mathbb{Z}_2$，或 D_4：长方形的对称群）。

Let G be a group of even order. Prove that G contains an element of order 2.

[Hint. Note that a non-identity element $x \in G$ has order 2 if and only if $x = x^{-1}$.]

We prove the contrapositive: if a finite group G has no element of order 2, then $|G|$ is odd.

For $x \neq e$: $x^2 = e \iff x = x^{-1}$, so x has order 2 if and only if $x = x^{-1}$. Assume no element has order 2. Then every $x \neq e$ satisfies $x \neq x^{-1}$, so $G \setminus \{e\}$ splits into pairs $\{x, x^{-1}\}$ of size 2. These pairs are disjoint: if $\{x, x^{-1}\}$ and $\{y, y^{-1}\}$ share an element, then by uniqueness of inverses they are the same pair. Hence

$$G = \{e\} \cup \{x_1, x_1^{-1}\} \cup \cdots \cup \{x_m, x_m^{-1}\} \quad (\text{disjoint}), \qquad |G| = 1 + 2m,$$

which is odd. Taking the contrapositive: if $|G|$ is even, G has an element of order 2.

> 板书画 G 中元素两两连线配对，e 单独一个点。常见错误：没说明各对互不相交（要用逆元唯一）；把"没有 2 阶元"理解成"没有自逆元"而漏掉 e。**注意：本题与 PS7 计分作业 Assigned Question (b)（D_6 中找两个 2 阶子群）相关——课上讲本题的配对论证没问题，但不要接着在 D_6 里列出具体的 2 阶元或 2 阶子群。**

Let $GL_2(\mathbb{R})$ be the group of 2×2 invertible matrices with entries in $\mathbb{R}$.

(a) For each positive integer n, find an element $A \in GL_2(\mathbb{R})$ of order n.

(b) Let A be an element of $GL_2(\mathbb{R})$ with $\det(A) \neq \pm 1$. Prove that A has infinite order.

(c) Find an element $A \in GL_2(\mathbb{R})$ such that $\det(A) = 1$ and A has infinite order.

(a) Let A be the anticlockwise rotation of $\mathbb{R}^2$ through $2\pi/n$:

$$A = \begin{pmatrix} \cos(2\pi/n) & -\sin(2\pi/n) \\ \sin(2\pi/n) & \cos(2\pi/n) \end{pmatrix}.$$

Then A^k is the rotation through $2\pi k/n$, which equals I_2 if and only if $n \mid k$. So the smallest such $k \in \mathbb{Z}^+$ is n, and $|A| = n$ (for $n = 1$, $A = I_2$).

(b) Since $\det(BC) = \det(B)\det(C)$, we have $\det(A^n) = \det(A)^n$. As $\det A$ is a non-zero real number different from ± 1, $|\det A| \neq 1$. If $|\det A| > 1$ then $|\det(A^n)| = |\det A|^n > 1$; if $0 < |\det A| < 1$ then $|\det A|^n < 1$. Either way $\det(A^n) \neq 1 = \det(I_2)$, so $A^n \neq I_2$ for every $n \in \mathbb{Z}^+$. Hence A has infinite order.

(c) Take $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, with $\det A = 1$. By induction, $A^n = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$: true for $n = 1$, and $A^{n+1} = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & n+1 \\ 0 & 1 \end{pmatrix}$. Since $n \neq 0$, $A^n \neq I_2$ for all $n \in \mathbb{Z}^+$, so $|A| = \infty$. (Another example: $\begin{pmatrix} 2 & 0 \\ 0 & 1/2 \end{pmatrix}$.)

> 给学得快的同学，三小问各一句话思路：旋转矩阵；行列式的幂；剪切矩阵。常见错误：(b) 只考虑 $\det A > 1$ 的情况，漏掉 $0 < |\det A| < 1$ 和负数；(c) 只算 A^2 就下结论，没有给出一般 A^n。追问：$\det A = -1$ 的元素可以有有限阶吗？（可以，如反射，阶 2）。

Let $G = S_n$ be the symmetric group of degree n, where $n \geqslant 2$. An element in G is called a *transposition* if it is a 2-cycle.

(a) Show that every element in G can be written as a product of transpositions.
[Hint. We have $(1,2,3,4) = (1,2)(2,3)(3,4)$.]

(b) Show that every transposition in G can be written as a product of an odd number of transpositions of the form $(i, i+1)$.
[Hint. We have $(1,3) = (1,2)(2,3)(1,2)$.]

(c) For each $g \in G$, define

$$N(g) = |\{(a,b) \in \{1,\ldots,n\} \times \{1,\ldots,n\} : a < b \text{ and } g(a) > g(b)\}|\,.$$

Show that $|N(g(i,i+1)) - N(g)| = 1$ for all $i \in \{1,\ldots,n-1\}$.

(d) For each $g \in G$ define

$$\operatorname{sgn}(g) = (-1)^{N(g)},$$

which we call the sign (or signature) of g. Using parts (a)–(c), deduce that $\operatorname{sgn}(g) = 1$ if g can be written as the product of an even number of transpositions, otherwise $\operatorname{sgn}(g) = -1$.

(e) Let $H = \{g \in G : \operatorname{sgn}(g) = 1\}$. Prove that H is a subgroup of G and calculate $|H|$.

(a) Every permutation is a product of disjoint cycles, and $e = (1,2)(1,2)$, so it suffices to treat a k-cycle ($k \geqslant 2$). We claim

$$(i_1, i_2, \ldots, i_k) = (i_1, i_2)(i_2, i_3)\cdots(i_{k-1}, i_k).$$

Apply the right-hand side to i_j, working right to left. For $j < k$: the factors to the right of (i_j, i_{j+1}) only move $i_{j+1}, \ldots, i_k$, so i_j is unchanged until (i_j, i_{j+1}) sends it to i_{j+1}; the factors further left only move $i_1, \ldots, i_j$, so it stays at i_{j+1}. For $j = k$: (i_{k-1}, i_k) sends $i_k \mapsto i_{k-1}$, then (i_{k-2}, i_{k-1}) sends it to i_{k-2}, and so on, ending at i_1. Points outside the cycle are fixed by every factor. So both sides agree.

(b) Call $(j, j+1)$ a *swap*. Every transposition is $(i, i+k)$ with $1 \leqslant k \leqslant n-1$. We prove by induction on k the statement $P(k)$: every $(i, i+k) \in S_n$ is a product of an odd number of swaps. $P(1)$ holds, since $(i, i+1)$ is itself a swap. Assume $P(k)$ for some $k \leqslant n-2$. Then

$$(i, i+k+1) = (i, i+1)\,(i+1, i+1+k)\,(i, i+1),$$

(check: $i \mapsto i+1 \mapsto i+1+k \mapsto i+1+k$; $i+1+k \mapsto i+1+k \mapsto i+1 \mapsto i$; $i+1 \mapsto i \mapsto i \mapsto i+1$; all else fixed). By $P(k)$, the middle factor is a product of an odd number m of swaps, so $(i, i+k+1)$ is a product of $m+2$ swaps, again odd. So $P(k+1)$ holds, and by induction $P(k)$ holds for all k.

(c) Let $f = g(i, i+1)$. Then $f(i) = g(i+1)$, $f(i+1) = g(i)$ and $f(c) = g(c)$ for $c \neq i, i+1$. Compare the pairs $a < b$ counted by $N(f)$ and $N(g)$:

- Pairs with $\{a,b\} \cap \{i,i+1\} = \varnothing$: same values, so counted by both or neither.
- For fixed $c < i$: the two pairs $(c,i),(c,i+1)$ are inverted by f exactly as $(c,i+1),(c,i)$ are inverted by g; so together they contribute the same amount to $N(f)$ and $N(g)$. The same holds for fixed $c > i+1$ and the pairs $(i,c),(i+1,c)$.
- The pair $(i,i+1)$: $f(i) > f(i+1) \iff g(i+1) > g(i) \iff g(i) < g(i+1)$, so it is counted by exactly one of $N(f)$, $N(g)$.

Hence $N(f) - N(g) = \pm 1$.

(d) Suppose $g = t_1 \cdots t_r$ with each t_j a transposition. By (b), each t_j is a product of an odd number of swaps, so $g = s_1 s_2 \cdots s_q$ is a product of q swaps with $q \equiv r \pmod 2$. Now $N(e) = 0$, and by (c) each time we multiply on the right by a swap, N changes by ± 1, so sgn is multiplied by -1. Building $g = e\, s_1 s_2 \cdots s_q$ one swap at a time gives $\operatorname{sgn}(g) = (-1)^q = (-1)^r$. So $\operatorname{sgn}(g) = 1$ if r is even and -1 if r is odd. (In particular the parity of r does not depend on how g is written.)

(e) $e = (1,2)(1,2)$, so $e \in H \neq \varnothing$. Let $g, h \in H$, written as products of r and s transpositions, with r, s even by (d). Since each transposition is its own inverse, $g^{-1} = (g_1 \cdots g_r)^{-1} = g_r \cdots g_1$, so $g^{-1}h$ is a product of $r+s$ transpositions, an even number. Hence $g^{-1}h \in H$, and $H \leqslant G$ by B5.

For the order, let $z = (1,2)$ and define $\psi : H \to G \setminus H$, $\psi(h) = zh$. If $h \in H$ is a product of an even number of transpositions, zh is a product of an odd number, so $zh \in G \setminus H$ by (d). ψ is injective by left cancellation, and surjective: for $g \in G \setminus H$, $zg \in H$ and $\psi(zg) = z^2 g = g$. Therefore $|H| = |G \setminus H|$, so $n! = |G| = 2|H|$ and $|H| = \frac{1}{2} n!$.

> 挑战题，只给学得快的学生，课上至多讲 (a) 的拆法和 (c) 的思路（看“逆序对”）。(a) 用 $(1,2,3) = (1,2)(2,3)$ 从右往左验算一次。**sol7 (c) 的说明只讨论了 $a \in \{i, i+1\}$、$b > i+1$ 的对，漏了 $b \in \{i, i+1\}$、$a < i$ 的对（同理成立），这里已补全。**常见错误：(d) 只说“每次乘一个对换符号变号”，但 (c) 只对相邻对换 $(i, i+1)$ 成立，必须先用 (b)。追问：$|H|$ 在 S_3 里是多少，分别是哪些元素？（3 个：e 和两个 3-循环）。

Sources and scope

Pure: Problem Sheet 7 with solutions (attached to Tim Burness's email "Intro to Pure: End of Week 7 updates", 6 Nov 2026); Blackboard "Lecture summary" read on 11 Nov 2026 (lectures of 9 and 10 Nov: Sections 7.4–7.5); announcement "Week 8 updates" (9 Nov 2026).

Linear: "Linear Algebra tutorial ideas 2026", Week 8 section (R. Carey, emailed 10 Nov 2026); Linear Algebra Problem Sheet 7 (Blackboard, read 11 Nov 2026); Charles Cox's announcement "Week 7 recap and coming up in week 8" (5 Nov 2026).

Hand-in questions are hints only: Pure PS7 Assigned Question (assessed, due 13:00 Thu 19 Nov), Pure PS6 Assigned Question (due 13:00 Thu 12 Nov), Linear PS7 Q1 (due noon Wed 18 Nov). The student handouts contain questions, key ideas and working space only — no answers.

> 本页给老师看（含答案），不要发给学生；学生只拿 Student PDF。